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Message Passing Interface Parallelization for Two-Way Coupled Euler–Lagrange Simulation of Microbubble Enhanced HIFU

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Microbubble enhanced high intensity focused ultrasound (HIFU) is of great interest to tissue ablation for tumor treatment such as in liver and brain cancers. To accurately characterize the acoustic and thermal fields during this process, a coupled Euler–Lagrange model is used. The ultrasound field is modeled using compressible Navier–Stokes equations on an Eulerian grid, while the microbubbles are tracked in a Lagrangian fashion. The coupling is realized through the void fraction computed from the instantaneous bubble volumes. To speed up the computations, an message passing interface parallelization scheme based on domain decomposition is herein proposed. During each time-step, message passing interface processors, each handling one subdomain, are first used to execute the fluid computation, and then the bubble computations. This is followed by the coupling procedure. The coupling is challenging as the effect of the bubbles through the void fraction at an Eulerian point near a subdomain border will require information from bubbles located in different subdomains, and vice versa. This is addressed by a special utilization of ghost cells surrounding each fluid subdomain, which allows bubbles to spread their void fraction effects across subdomain edges without the need of exchanging directly bubble information between subdomains and significantly increasing overhead. After a careful verification of gas effects conservation, this parallelization scheme is validated and illustrated on a typical microbubble enhanced HIFU problem, followed by parallelization scaling tests and efficiency analysis. [DOI: 10.1115/1.4051148]

Introduction

High intensity focused ultrasound (HIFU) is applied in therapeutic and surgical medical interventions such as tissue ablation in cancer treatment. This is particularly the case for deep-seated tumors such as in the liver and the brain [1,2] because HIFU is a truly noninvasive form of localized ablative therapy. To widen the clinical adoption of HIFU treatment, one promising technique is to inject ultrasonic contrast agents in the target region [3,4] in order to enhance the reaction. The interaction between the ultrasonic contrast agents and HIFU increases target tissue heating via bubble dynamics. This allows the use of lower intensity HIFU and lowering of the risk of undesirable prefocal damage. Further understanding of microbubble-assisted HIFU acoustic and thermal fields are essential to predicting the bio-effects of HIFU on the target tissue and its surroundings and to develop standards to ensure the safety and efficacy of the treatments.

For positive results, it is highly desirable to use a computational tool, which could enable accurate characterization of the acoustic and thermal fields for bubble enhanced HIFU in clinically relevant environments. For this purpose, we have developed a fully compressible Eulerian–Lagrangian approach which couples the acoustic field with bubble dynamics [5,6]. Similarly, as in other multiphase flow applications such as in Refs. [7–12], in HIFU studies the Eulerian–Lagrangian method has shown intrinsic advantages owing to its capability to account for complex physics of nonlinear interaction between oscillating bubbles and ultrasound in the computation of heat deposition. However, owing to high computational costs in real applications, efficient parallelization schemes for significant speedup are critical.

In this paper, we present an message passing interface (MPI) parallelization scheme based on domain decomposition for both the continuum fluid and the discrete bubbles to take advantage of MPI scalability due to distributed memory features. The Eulerian computational domain is subdivided into several subdomains each having the same number of grids, and subsequently, the Lagrangian bubbles are subdivided based on their locations in each subdomain. The MPI realization for the fluid and bubble computations is relatively straightforward and follows routine MPI procedures. However, the key coupling step, which requires mapping the Lagrangian bubble solutions into the fixed Eulerian grids is challenging. This is due to the fact that we need to spread smoothly the effects of each bubble to all its adjacent computational cells with some potentially belonging to other subdomains. To overcome this, we introduce the concept of ghost cells surrounding each fluid subdomain as described further below.

The paper is organized as follows. We first summarize the main features of the two-way coupled, fully compressible Euler–Lagrange model. This is followed by a description of the MPI parallelization scheme, particularly focusing on Euler–Lagrange Mapping. After a brief discussion of simulation results for a bubble enhanced HIFU problem [5,6], scaling tests and efficiency analysis of the parallelization scheme are conducted and discussed. Finally, a summary of the findings is presented.

Eulerian–Lagrangian Model

The three-dimensional (3D) Eulerian–Lagrangian fully compressible model employed in this study has been documented and validated in our previous studies [5,6]. The present paper aims at accelerating the computations through MPI parallelization and does not include additional model development. Below we

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describe the key features of the Eulerian–Lagrangian model and then concentrate on the parallelization aspects.

Eulerian Compressible Mixture Flow Solver. The unsteady compressible flow solver for acoustic wave propagation through a two-phase medium and other associated physics includes mainly the governing equations for conservation of mass, momentum, and energy as follows:

$$\begin{aligned}&\frac{\partial \rho_m}{\partial t}+\nabla\cdot(\rho m\,\mathbf{u})=0\\&\frac{\partial \rho_m \mathbf{u}}{\partial t}+\nabla\cdot(\rho_m \mathbf{u}\mathbf{u}+\rho_m \mathbf{I}+\sigma)=0\\&\frac{\partial \rho_m E_m}{\partial t}+\nabla\cdot((\rho_m E_m+\rho_m)\mathbf{u})+\boldsymbol{\sigma}\cdot\mathbf{u}+q)\end{aligned} \tag{1}$$

Here the subscript $_m$ refers to the mixture properties, ρ_m is the mixture density, E_m and p_m are, respectively, the mixture total energy and pressure, $\mathbf{u}$ is the mixture velocity. The mixture density can be related to the volume fraction α of each component (such as tissue/phantom medium and the gas inside the bubbles) by

$$\rho_m=\sum_i \rho_i\alpha_i,\ \text{where}\ \sum_i \alpha_i=1 \tag{2}$$

Also, p_m and E_m are related to ρ_m with a stiffened equation of state [13].

The numerical scheme uses a fully conservative higher-order scheme based on MUSCL method and an approximate Riemann solver [14,15].

Lagrangian Modeling of Dispersed Phase. For the bubbles dispersed in the medium, we use a Lagrangian bubble tracking method. All bubbles are treated as point sources to account for volume changes and dipoles to account for translation. The bubble volume variations are obtained using a modified Rayleigh–Plesset–Keller–Herring equation [16]

$$\begin{aligned}&\left(1-\frac{\dot R}{c_m}\right)R\ddot R+\frac{3}{2}\left(1-\frac{\dot R}{3c_m}\right)\dot R^2=\frac{(\mathbf{u}-\mathbf{u}_b)^2}{4}\\&+\frac{1}{\rho_m}\left(1+\frac{\dot R}{c_m}+\frac{R}{c_m}\frac{d}{dt}\right)\begin{bmatrix}p_v+p_g-p_m-\dfrac{2\gamma}{R}\\4\mu_m\dfrac{\dot R}{R}-\dfrac{4}{3}G\left[1-\left(\dfrac{R_0}{R}\right)^3\right]\end{bmatrix}\end{aligned} \tag{3}$$

In the above equation, R is the bubble radius and the dots represent time differentiation. μ_m is the mixture dynamic viscosity and c_m is the local mixture sound speed. $\mathbf{u}_b$ is the bubble velocity, p_g is the gas pressure inside the bubble, and p_v is the liquid–vapor pressure. γ is the surface tension and G is the shear elasticity of the medium surrounding the bubble.

The bubble trajectory is obtained using the motion equation of Johnson and Hsieh [17]

$$\begin{aligned}\frac{d\mathbf{u}_b}{dt}=&-\frac{3}{\rho_m}\nabla p+\frac{3}{4}\frac{C_D}{R}(\mathbf{u}-\mathbf{u}_b)|\mathbf{u}-\mathbf{u}_b|\\&+\frac{3}{2\pi}\frac{C_L}{R}\sqrt{\frac{\mu_m}{\rho}}\frac{(\mathbf{u}-\mathbf{u}_b)\times\boldsymbol{\omega}}{\sqrt{|\boldsymbol{\omega}|}}+\frac{3}{R}(\mathbf{u}-\mathbf{u}_b)\dot R\end{aligned} \tag{4}$$

where C_D is the drag coefficient, C_L is the lift coefficient, and ω is the local vorticity.

Eulerian–Lagrangian Coupling. The two-way coupling between the Eulerian continuum-based model and the Lagrangian discrete bubble model is realized as follows [18–21]:

- The mixture flow field has an evolving mixture density in space and time which satisfies mass, momentum, and energy conservation,
- The bubble dynamics and the motion of the individual bubbles in the flow field are controlled by the two-phase medium's local properties and gradients,
- The local properties of the mixture are determined by the bubble sizes and their space distribution.

The key of this coupling procedure is to map to the fixed Eulerian grids the Lagrangian bubbles solutions, specifically, to derive the void fraction in each computational cell from the instantaneous bubble sizes and locations. To do so efficiently, as illustrated in Fig. 1, a Gaussian distribution scheme is employed to smoothly "spread" each bubble volume effect over all neighboring cells [18–20]. The void fraction is computed using the following formulation:

$$\alpha_i=\sum_{j=1}^{N_i}\frac{f_{i,j}V_j^b}{\sum_k^{N_{\text{cells}}} f_{k,j}V_k^{\text{cell}}} \tag{5}$$

where V_j^b is the volumes of bubble, j, and V_k^{cell} is the volume of cell, k, respectively. N_i is the number of bubbles which are in the "influence range" of a cell i. N_{cells} is the number of cells "influenced" by a bubble j, and $f_{i,j}$, is a Gaussian distribution weight factor (for bubble j contribution to cell i). The contribution gradually decays to zero as the distance between the bubble and a given cell increases.

Modeling of Heat Deposition. Once the multiphase flow field solution is obtained from the above-described approach, heat source terms are derived: $q_{\text{US,AC}}$ due to ultrasound absorption and acoustic emission, and q_{VIS}, due to viscous damping of the bubbles

$$q_{\text{US,AC}}=\mu_b\varepsilon_{kk}^2+2\mu\left(\varepsilon_{ij}-\frac{1}{3}\varepsilon_{kk}\delta_{ij}\right)^2 \tag{6}$$

where μ and μ_b are the shear and bulk viscosities of the fluid and ε is the strain rate tensor [6]. q_{VIS} is computed through a procedure very similar to that of void fraction computation described in Eq. (5), which sums up the smoothly spread contributions from each bubble within the "influence range"

$$\begin{aligned}q_{\text{VIS}}&=\sum_{j=1}^{N_i}\frac{q_{\text{vis}}^b V_j^b}{\sum_{k=1}^{N_{\text{cells}}} v_{k,j}V_k^{\text{cell}}}\\q_{\text{VIS}}^b&=(4\pi R^2)\cdot\left(4\mu\frac{\dot R^2}{R}\right)\end{aligned} \tag{7}$$

A heat transfer equation is then resolved to compute the heat deposition effect using the heat sources $q_{\text{US,AC}}$ and q_{VIS}

$$\rho C_p\frac{\partial T}{\partial \tau}=K\nabla^2 T+q_{\text{US,AUS}}+q_{\text{VIS}} \tag{8}$$

where ρ, C_p, and K are the density, specific heat, and thermal conductivity of the medium, T the temperature, and τ the time.

Parallelization Algorithm and Implementation

Algorithm and Overall Procedure. As stated previously, the Eulerian computational domain is subdivided into several subdomains, as illustrated in Fig. 2. The number of subdomains is matched with the number of processors in the available distributed-memory system, or as required by the user. During

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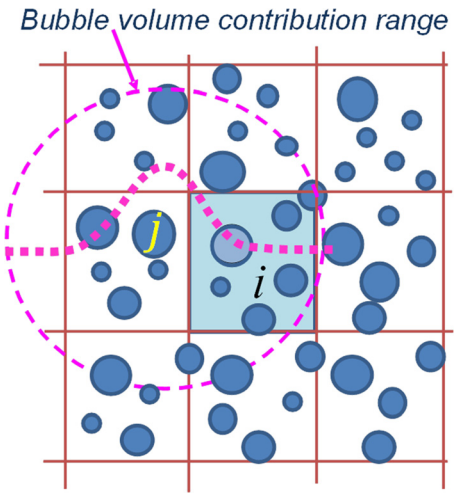


Fig. 1 Sketch of the void fraction computation using the Gaussian distribution scheme [18]

every computation time-step, the involved MPI processors handle each one subdomain and operate simultaneously to:

Step 1. Solve Eqs. (1) and (2) for fluid computation,
Step 2. Solve Eqs. (3) and (4) for all bubbles,
Step 3. Map the Lagrangian bubble solutions into the Eulerian grids of each subdomain using (5).

These operations proceed for each physical time-step until the required final time is reached.

Note that in MPI parallelization, each processor operates independently except when communication with other processors is needed. Therefore, Steps 1 and 2 are relatively straightforward and are very similar to serial computations. Interprocessor communication is necessary where cells near a subdomain edge require information from neighboring subdomains handled by different processors. This can be realized through the addition of ghost cells along the borders of each subdomain. As illustrated in Fig. 3, each subdomain is surrounded by N layers of ghost cells along its edges, with N depending of the order of the numerical scheme.

Figure 4 gives further details on how ghost cells are used to realize communication between subdomain A and its neighbor B. As seen in the figure, the boundary data in the real cells of B are copied to fill the corresponding A ghost cells, which share the same geometric location. At the same time, the parallel operation is performed in the opposite direction, i.e., the boundary data from subdomain A is used to fill the corresponding ghost cells data in B. This two-way data movement between neighboring subdomains could be easily realized by routinely calling standard MPI library functions, e.g., MPI_SENDRECV.

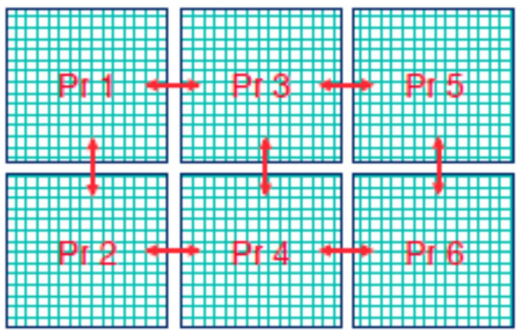


Fig. 2 Illustration of the computation domain decomposition

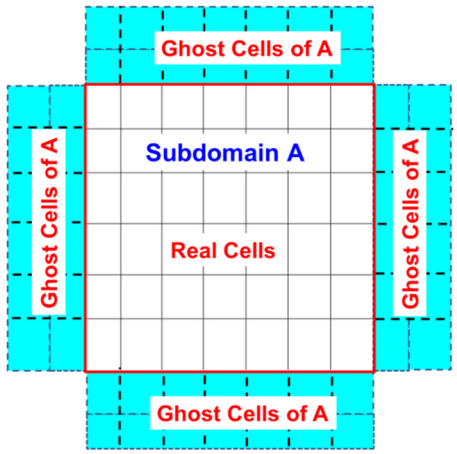


Fig. 3 Definition of N layers of "ghost cells" surrounding a subdomain (here $N = 2$). The red lines indicate the border of subdomain. The white cells inside subdomain are "real" cells. Those in blue along each border line are "ghost" cells. (Color version online.)

However, Step 3 is more challenging as the effect of the bubbles through the void fraction at an Eulerian point near a subdomain borders will require information from bubbles located in different subdomains. Similarly, a bubble near subdomain borders has to spread its void fraction contribution to cells belonging to different subdomains. Therefore, we mainly focus below on addressing how to map Lagrangian bubble solutions into the Eulerian framework in the MPI environment.

Parallelization of Euler–Lagrange Mapping. As stated above, each subdomain is surrounded by N layers of ghost cells along its borders. Traditionally, ghost cells do not participate in the computations but are merely used to create a transfer zone to load data from the "real" cells along the edge of a neighboring subdomain. This means the data movement direction is: *Real* → *Ghost*, as indicated by the direction of the orange arrows in Fig. 4. However, to enable mapping in the Eulerian–Lagrangian scheme, we explored a different way of using the ghost cells. Here, the ghost cells participate in the computations and then copy their values to the "real" cells along the edge of the neighboring subdomain (i.e., *Ghost* → *Real*).

As shown in the example in Fig. 5, a bubble located close to the right edge of subdomain A spreads its void contribution into the neighboring subdomain B, immediately to its right. First, we smoothly spread the bubble's void effects into all adjacent cells regardless of whether they are real or ghost cells. Then, using MPI

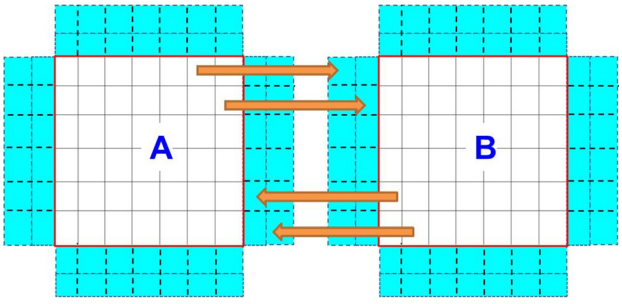


Fig. 4 Sketch of data copying from real cells of Subdomain B to the ghost cells of subdomain A and vice versa from real cells of subdomain A to the ghost cells of subdomain B. (Note that the right edge of the white real parts of A and the left edge of B are actually superposed but have been separated here to illustrate the procedure).

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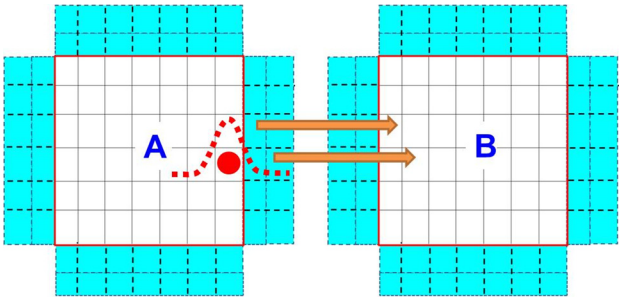


Fig. 5 Illustration of the void fraction contribution for a bubble (red sphere) near the edge of a subdomain A spreading its contribution into its neighboring subdomain B. The blue cells are ghost cells of A. The orange arrows indicate the "void" effects are copied from "ghost" cells of A and added into "real" cells along the left edge of B. (Color version online.)

communication, the void values in the A ghost cells are transferred to the neighboring B subdomain and added into the corresponding real edge cells. (Note that now the direction of orange arrows in Fig. 5 is opposite to those in Fig. 4). By doing so, the bubbles near the edge are able to spread void effects across subdomain edges without the need of sending bubble information to other subdomains. Vice versa, the opposite occurs, i.e., a bubble near the left edge of subdomain B spreads its effects to subdomain A in the same way.

The next example, shown in Fig. 6, is for a more complex condition, where a bubble is located near the corner of a subdomain A, which has neighboring subdomains in two directions (i.e., here top and right). Here again, the bubble first smoothly spreads its void effects into all adjacent cells no matter if they are real (in white), ghost (in blue), or doubly ghost (in green). Next, MPI communication is conducted direction by direction. As illustrated in the figure, the following operations are conducted:

(1) First, void fraction values are copied from the doubly ghost cells of subdomain A and added into the ghost cells along the top edge of the right subdomain B.
(2) Then, the updated values in the ghost cells of B are further added into the real cells along the bottom edge of subdomain C.

The examples are shown for simplicity in two-dimensional configurations. The same logic applies to three-dimensional configurations.

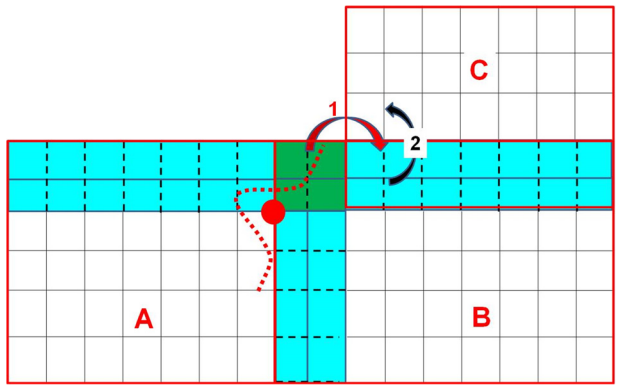


Fig. 6 Illustration of the void fraction contribution for a bubble near the corner of a subdomain A. The cells in green are "doubly ghost" cells used to communicate with the top-right corner neighbor subdomain C. Arrow 1 in red indicates the void effects are copied from "doubly ghost" cells and added into "ghost" cells belonging to right subdomain B. Arrow 2 in black means the void effects are then copied from the ghost cells of B and further added into the real cells of subdomain C. (Color version online.)

The successful implementation of this unconventional way of the utilization of ghost cells for void fraction computation is verified in Fig. 7. A bubble located very close to the top right corner of subdomain A is shown to be able to smoothly spread its void fraction into its subdomain A and the three neighboring subdomains B, C, and D.

For further validation purposes, we also consider the case where a spherical bubble cloud located at the center of a 3D cubic fluid domain is excited by pressure waves, similar to our previous study in Ref. [21]. Herein, 5000 5 μm radius bubbles are randomly distributed inside a spherical region of radius 5 mm. Figure 8 illustrates the decomposition of the 3D domain into subdomains in the three directions. As seen in Fig. 9(*a*), the resulting void fraction contours are smoothly continuous across all subdomain edges. More importantly, further, very good quantitative validation is obtained in Fig. 9(*b*) by comparing the time evolution of the total gas volume in the cloud obtained either by summing at each time-step the volume of all bubbles in the cloud or by integrating at each time-step the void fraction in all the fluid cells.

As stated earlier, other bubble effects such as thermal effects are handled in the same way as the void fraction computations.

Test Simulation and Results

In this section, we apply the two-way coupled Eulerian–Lagrangian approach to simulate HIFU enhancement using microbubbles. We consider here a HIFU configuration from our earlier Euler–Lagrange model implementation in Refs. [5] and [6], results of which we had successfully compared with in vitro experiments [4], and now examine how much the newly developed MPI parallelization can speed up the solution. In this problem, a 2.2 MHz transducer was used to insonate a phantom tissue made of Polyacrylamide gel in the presence of bubbles [4]. Phantom tissue was used for its relative ease of implementation in a controlled laboratory setting. The phantom tissue volume was cylindrical, with a radius of 25 mm and a height of 60 mm.

In our computations, as in Refs. [5] and [6], the cylindrical (x, θ, z) computational domain of physical dimensions of 29.5 mm $\times$ 360 deg $\times$ 60 mm, is discretized using $N_x = 608 \times N_\theta = 720 \times N_Z = 1415$ cells. A cloud of bubbles of radius 1.3 μm is uniformly distributed in a smaller cylindrical volume of diameter 10 mm and length 10 mm and centered at the geometric focus of the excitation transducer ($z = 40$ mm).

To take advantage of the axisymmetry of the problem, we used only 1 cell in the θ-direction, i.e., reduced N_θ to 1 and extended the solution to the full 360 deg domain. The bubble cloud was excited by an HIFU pressure wave emitted from the plane $z = 0$ mm, with a frequency of 2.2 MHz and an amplitude of 3.0 atm. The wave travels through the phantom tissue and reaches

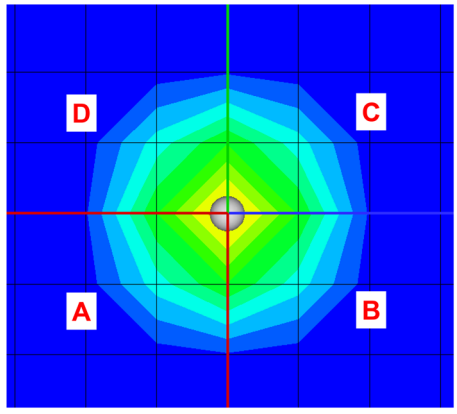


Fig. 7 Void fraction contour across four neighboring subdomains A, B, C, and D resulting from a single bubble very close to the top-right corner of the bottom-left subdomain A

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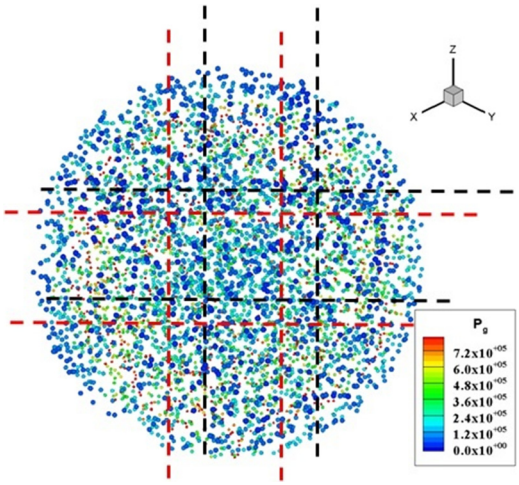
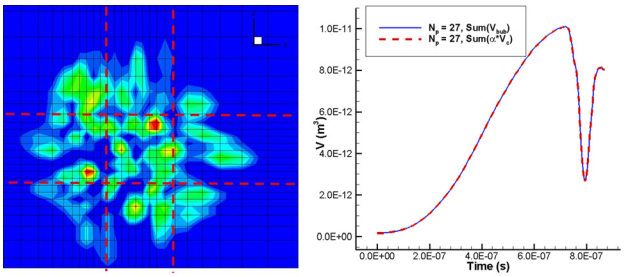


Fig. 8 Illustration of domain decomposition cutting through a spherical bubble cloud



(*a*) (*b*)

Fig. 9 (*a*) Void fraction contour on the central plane of a spherical bubble cloud and (*b*) comparison of time history of the total gas volume computed by summing bubble volume and integrating void fraction through all cells

the bottom of the cloud. As seen in Fig. 10, the simulation captures the scattering of the pressure waves by the bubbles and their attenuation before they reach the focus.

The obtained temperature history near the focus is shown in Fig. 11 for three bubble cloud void fractions, $\alpha = 0$, $\alpha = 10^{-5}$, and $\alpha = 10^{-4}$. The latter two result in a temperature rise relative to the $\alpha = 0$ condition of about 5 K and 15 K, respectively. Comparisons between the 3DynaFS numerical results and the experiments are reasonable for $\alpha = 0$ and $\alpha = 10^{-4}$. However, there is a noticeable deviation at $\alpha = 10^{-5}$. These may be attributed to probably different distribution of the bubbles at this low void fraction between the computations and the experiment. Also, notable is the uncertainty in the location where the temperature was measured in the experiments [4], which we tried to compensate for in Refs. [5] and [6] by considering different locations above and below the supposed focus point.

Below we focus on the main objective of this paper, i.e., parallelization efficiency tests and analysis.

Scaling Tests and Speedup Analysis. To examine the scalability of the developed scheme, we tested the code for bubble enhanced HIFU simulations on Dyna-Hive, one of Dynaflow's multinode, multiprocessor Linux clusters. Dyna-Hive consists of ten nodes, with each equipped with twelve 3G Xeon CPUs and 27 GB memory. Before presenting the scaling tests, the accuracy of the parallelization is examined in Fig. 12. The pressure history is monitored at a selected field point near the focal region inside the bubble cloud for different selections of the number of processors. Also, the histories of the radius of a selected bubble also near the focus are compared for different numbers of processors. As seen in Fig. 12, for both comparisons, the results are independent of the number of processors used.

Figure 13 shows the wall time per time-step spent by the three main modules in the coupled computations while Fig. 14 shows the corresponding speed-up factors for an increasing number of MPI processors. In these figures, each line corresponds to one main part of the computation, i.e., Eulerian, Lagrangian and coupling through the Eulerian–Lagrangian mapping. The Eulerian step is the most time-consuming in this case, while the Lagrangian computations and the coupling both are an order of magnitude faster.

Concerning parallelization, it is seen that the wall time per time-step for the Eulerian module (the most time-consuming part) drops from 15 s to less than 0.15 s as the processor number increases from 1 to 120, indicating a two-order of magnitude speedup. The speed-up factor in Fig. 14 is almost linear with increasing the number of processors except for some deviation for a few points. This is understandable and expected as the cells are evenly distributed when the fluid domain is decomposed into multiple subdomains. Similar results are seen for the Lagrangian part of computation.

However, surprisingly, the speed-up of the Eulerian–Lagrangian coupling through mapping performs even better; a nearly 240 times speedup is achieved when 120 processors are

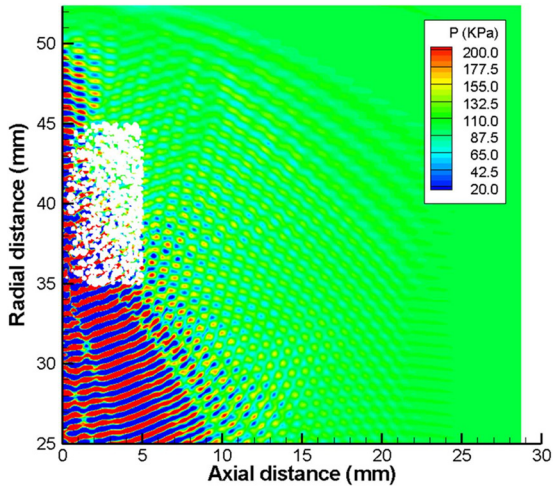


Fig. 10 Instantaneous pressure contours from HIFU simulations in phantom tissue with bubbles in a 10 mm × 10 mm cylindrical domain

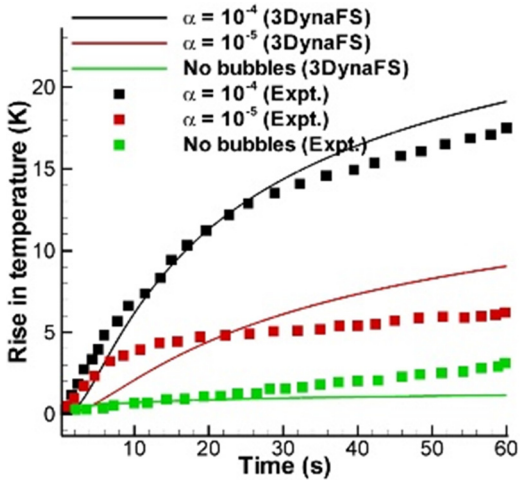


Fig. 11 Comparison of the temperature history with the experimental data of [4] near the focus [5,6]

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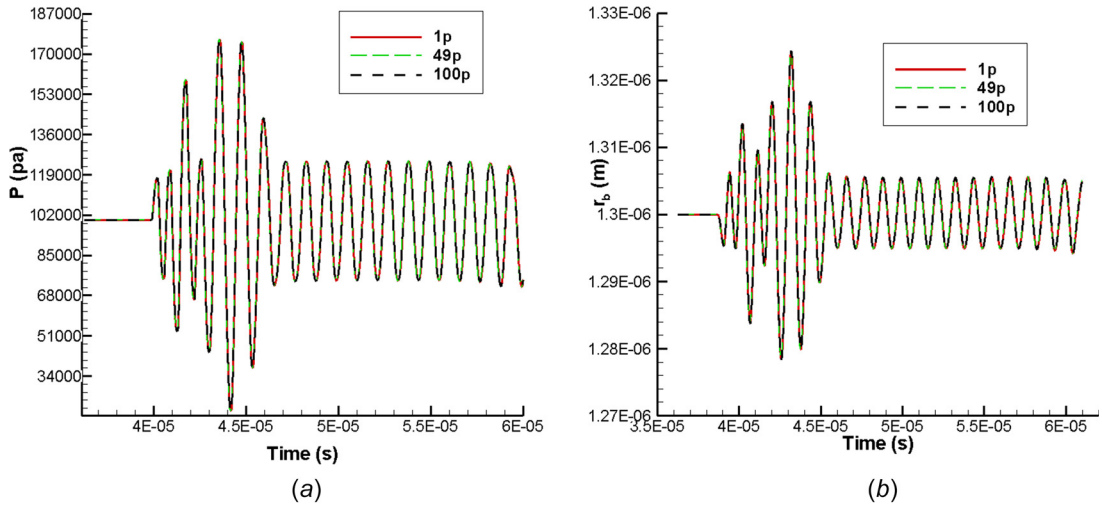


Fig. 12 Comparison of the time history of (*a*) the pressure at a fixed point for different MPI processors being used and (*b*) bubble size for a selected bubble inside the bubble cloud also for different number of processors

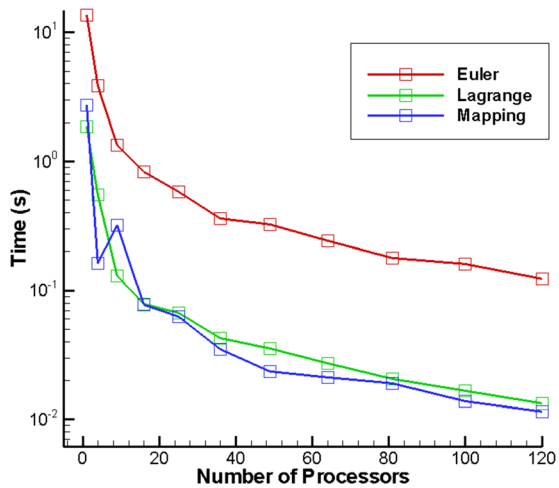


Fig. 13 Wall time per step for bubble enhanced HIFU simulation

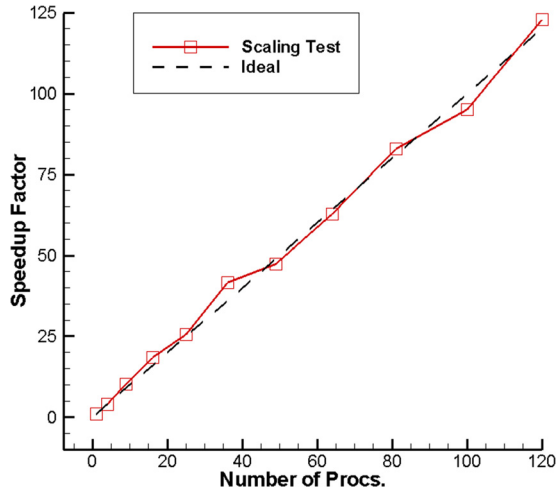


Fig. 15 Speed-up factor of total computational time per time-step using 1–120 MPI processors

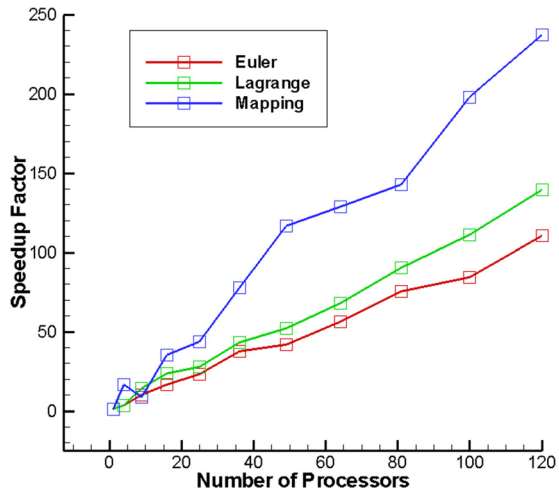


Fig. 14 Speed-up factors for different numbers of processors

used. This indicates that the newly introduced ghost cells procedure described above involves minimal overhead since it short-circuits sending bubbles across subdomains as in traditional parallelization approaches. The new ghost cells approach transforms the MPI communication to an operation similar to the parallelization of the Eulerian part. In addition, since the mapping involves the interaction between cells and bubbles, the numbers of both are reduced when the number of decomposed subdomains increases. This also explains why the parallelization of the Eulerian– Lagrangian mapping part is better than a linear speedup.

Figure 15 displays the speedup factor in terms of the total computational time per step summing the cost for all three parts. Owing to the good performance of each of these three parts presented above, an excellent speedup effect of our MPI parallelization is observed. In another word, this indicates at least a two-order of magnitude of speedup can be achieved just using our small in-house cluster. This is particularly helpful to carry out parametric studies of bubble enhancement of HIFU to optimize experimental designs or to plan clinical research.

Conclusion

An MPI parallelized 3D Eulerian–Lagrangian model was developed to study the enhancement of HIFU with microbubbles. This

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parallelization scheme adopts domain decomposition for both the continuum Eulerian domain and the Lagrangian discrete bubbles. An innovative utilization of ghost cells surrounding each subdomain successfully overcomes the difficulties encountered in mapping the Eulerian and Lagrangian solutions. This new scheme avoids the need to sending bubble information across all processors and enables the bubbles to smoothly spread their effects into all neighboring subdomains.

After careful verifications, efficiency tests were conducted for a bubble enhanced HIFU problem. More than a two-orders of magnitude speedup could be achieved using 120 processors on a Linux cluster consisting of ten 12-core Intel Xeon Nodes.

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References

[1] Kennedy, J. E., Wu, F., Ter Haar, G. R., Gleeson, F. V., Phillips, R. R., Middleton, M. R., and Cranston, D., 2004, "High-Intensity Focused Ultrasound for the Treatment of Liver Tumours," Ultrasonics, **42**(1–9), pp. 931–935.

[2] Hynynen, K., and Clement, G., 2007, "Clinical Applications of Focused Ultrasound—The Brain," Int. J. Hyperthermia, **23**(2), pp. 193–202.

[3] Tran, B. C., Hall, T. L., Fowlkes, J. B., and Cain, C. A., 2003, "Microbubble-Enhanced Cavitation for Noninvasive Ultrasound Surgery," IEEE Trans. Ultrason., Ferroelectr., Freq. Control, **50**(10), pp. 1296–1304.

[4] Okita, K., Sugiyama, K., Takagi, S., and Matsumto, Y., 2013, "Microbubble Behavior in an Ultrasound Field for High Intensity Focused Ultrasound Therapy Enhancement," J. Acoust. Soc. Am., **134**(2), pp. 1576–1585.

[5] Gnanaskandan, A., Hsiao, C.-T., and Chahine, G., 2018, "Numerical Simulation of High Intensity Focused Ultrasound (HIFU) Using A Fully Compressible Multiscale Model," 10th International Symposium on Cavitation – CAV2018, Baltimore, MD, May 14–16, Paper No. CAV18-05158.

[6] Gnanaskandan, A., Hsiao, C.-T., and Chahine, G., 2019, "Modeling of Microbubble-Enhanced High-Intensity Focused Ultrasound," Ultrasound Med. Biol., **45**(7), pp. 1743–1761.

[7] Farrell, K. J., 2003, "Eulerian/Lagrangian Analysis for the Prediction of Cavitation Inception," ASME J. Fluids Eng., **125**(1), pp. 46–52.

[8] Hsiao, C.-T., and Chahine, G., 2005, "Scaling of Tip Vortex Cavitation Inception Noise With a Bubble Dynamics Model Accounting for Nuclei Size Distribution," ASME J. Fluids Eng., **127**(1), pp. 55–65.

[9] Dinsmore, C., Aminfar, A., and Princevac, M., 2017, "Dissipative Effects of Bubbles and Particles in Shear Flows," ASME J. Fluids Eng., **139**(6), p. 061302.

[10] Kinzel, M. P., Lindau, J. W., and Kunz, R. F., 2019, "An Assessment of Computational Fluid Dynamics Cavitation Models Using Bubble Growth Theory and Bubble Transport Modeling," ASME J. Fluids Eng., **141**(4), p. 041301.

[11] Khojasteh-Manesh, M., and Mahdi, M., 2019, "Evaluation of Cavitation Erosion Intensity in a Microscale Nozzle Using Eulerian–Lagrangian Bubble Dynamic Simulation," ASME J. Fluids Eng., **141**(6), p. 061303.

[12] Melissaris, T., Bulten, N., and Van Terwisga, T. J. C., 2019, "On the Applicability of Cavitation Erosion Risk Models With a URANS Solver," ASME. J. Fluids Eng., **141**(10), p. 101104.

[13] Pelanti, M., and Shyue, K.-M., 2014, "A Mixture-Energy-Consistent Six-Equation Two-Phase Numerical Model for Fluids With Interfaces, Cavitation and Evaporation Waves," J. Comput. Phys., **259**, pp. 331–357.

[14] Colella, P., 1985, "A Direct Eulerian MUSCL Scheme for Gas Dynamics," SIAM J. Sci. Stat. Comput., **6**(1), pp. 104–117.

[15] Kapahi, A., Hsiao, C.-T., and Chahine, G., 2015, "A Multi-Material Flow Solver for High Speed Compressible Flows," Comput. Fluids, **115**, pp. 25–45.

[16] Plesset, M. S., and Prosperetti, A., 1977, "Bubble Dynamics and Cavitation," Annu. Rev. Fluid Mech., **9**(1), pp. 145–185.

[17] Johnson, V. E., and Hsieh, T., 1966, "The Influence of the Trajectories of Gas Nuclei on Cavitation Inception," Sixth Symposium on Naval Hydrodynamics, Washington, DC, Sept. 28–Oct. 4, pp. 163–179.

[18] Ma, J., Hsiao, C.-T., and Chahine, G. L., 2015, "Spherical Bubble Dynamics in a Bubbly Medium Using an Euler–Lagrange Model," Chem. Eng. Sci., **128**, pp. 64–81.

[19] Ma, J., Hsiao, C.-T., and Chahine, G. L., 2015, "Shared-Memory Parallelization for Two-Way Coupled Euler-Lagrange Modeling of Cavitating Bubbly Flows," ASME J. Fluids Eng., **137**(12), p. 121106.

[20] Ma, J., Hsiao, C.-T., and Chahine, G. L., 2015, "Euler-Lagrange Simulations of Bubble Cloud Dynamics Near a Wall," ASME J. Fluids Eng., **137**(4), p. 041301.

[21] Ma, J., Hsiao, C.-T., and Chahine, G. L., 2018, "Numerical Study of Acoustically Driven Bubble Cloud Dynamics Near a Rigid Wall," Ultrason. Sonochem., **40**, pp. 944–954.

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